

### 2.1.6 River Mouth Area and the Port

As mentioned earlier in sub-section 2.1.4, there are two ports in the mouth of the Itajaí River. They are the Itajaí Port at the right bank and the Portonave at the left bank. According to Itajaí Port, annual dredging of 3 million m<sup>3</sup> of the deposited sediments (fine sand, silt and clay) in the Port is necessary to maintain the water depth at 11 m for port function. On the other hand, the riverbed in the Port was seriously eroded due to the 2008 year flood, and thus the pier walls were destroyed (14 m deep to the pier foundation). The flow velocity in the Port was recorded 8.7 m/s during the 2008 year flood. At the river mouth, a jetty was installed to avoid blockage of river mouth due to sedimentation and groins (spur dike) are also installed to increase flow velocity by means of decreasing the flow area for controlling sedimentation in the mouth.



**Itajai Port (left), Navegantes Port (right)**



**Jetty and Spur dyke in River Mouth**

## 2.2 Flood Control Dams

The location map of 3 flood control dams is shown in Figure 2.1.1 and their main features are summarized in Table 2.2.1. The total drainage area of these 3 dams is 4,633km<sup>2</sup>, which is equivalent to 31% of the total drainage area of the Itajaí River mainstream. The flood control capacity of three dams is evaluated in terms of the equivalent rainfall to be stored (= reservoir storage capacity/drainage area). The estimated equivalent rainfall of the Norte dam shows the highest value of 154 mm. This means that the Norte dam has the largest flood control capacity among the dams compared to its drainage area and thus the likelihood of becoming full of water in reservoir and resulting in overflowing through spillway due to heavy rainfall might be low.

On the other hand, the estimated equivalent rainfalls of the Oeste and Sul Dams are 80 mm and 73 mm, which are almost half value of the Norte dam. Therefore, the reservoirs of these two dams easily become full. Especially the Oeste Dam has been full of water due to the floods in 2001 and 2010, causing overflowing through spillway, although no overflowing occurred at the Sul Dam. This might be due to more heavy rainfall occurred during the floods in the upper catchment of the Oeste Dam and smaller releasing capacity compared to that at the Sul Dam.

**Table 2.2.1 Main Features of Flood Control Dams**

|           |                        | <b>Oeste Dam</b>          | <b>Sul Dam</b>            | <b>Norte Dam</b>           |
|-----------|------------------------|---------------------------|---------------------------|----------------------------|
| Reservoir | Catchment area         | 1,042 km <sup>2</sup>     | 1,273 km <sup>2</sup>     | 2,318 km <sup>2</sup>      |
|           | Reservoir area         | 950 ha                    | 840 ha                    | 1,400 ha                   |
|           | Gross storage capacity | 83,000,000 m <sup>3</sup> | 93,500,000 m <sup>3</sup> | 357,000,000 m <sup>3</sup> |
|           | Equivalent rainfall    | 80 mm                     | 73 mm                     | 154 mm                     |
|           | Minimum water level    | EL.340.0 m                | EL.372.9 m                | EL.257.0 m                 |
|           | Maximum water level    | EL.363.0 m                | EL.408.0 m                | EL.304.25 m                |
| Dam       | Completed year         | 1973                      | 1976                      | 1992                       |
|           | Dam type               | Gravity                   | Rock fill                 | Anchoring                  |
|           | Dam height             | 25.0 m                    | 43.5 m                    | 58.5 m                     |

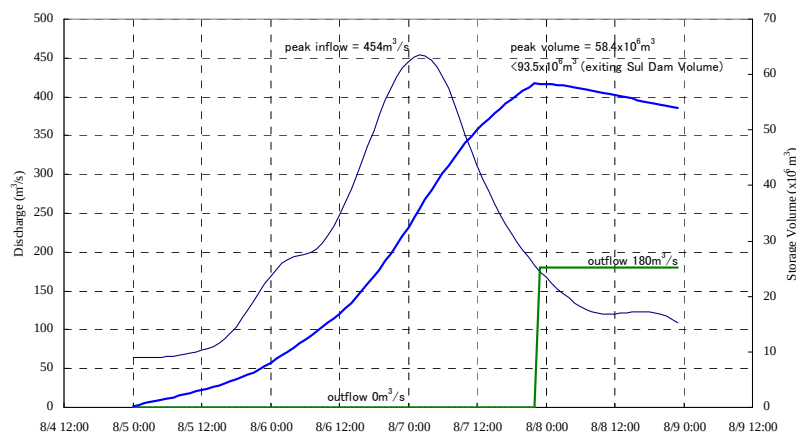
**Table 5.4.1 Initial Rainfall Loss due to Water Storage in Paddy Field by Sub-basin**

| Sub Basin       |                | Catchment Area<br>km <sup>2</sup> | Municipality       | Paddy Area<br>ha | Depth<br>cm | Rate of using<br>Rainwater | Storage<br>m <sup>3</sup> |            | Initial Loss<br>mm |
|-----------------|----------------|-----------------------------------|--------------------|------------------|-------------|----------------------------|---------------------------|------------|--------------------|
| Itadai do Oeste | Barragem Oeste | 1042                              | Rio do Campo       | 1,800            | 20          | 0.8                        | 2,880,000                 | 2,880,000  | 2.8                |
|                 | Basin 1        | 528                               | Mirim Doce         | 1,850            | 20          | 0.8                        | 2,960,000                 | 6,960,000  | 13.2               |
|                 |                |                                   | Taio               | 2,400            | 20          | 0.8                        | 3,840,000                 |            |                    |
|                 |                |                                   | Salete             | 100              | 20          | 0.8                        | 160,000                   |            |                    |
|                 | Basin 2        | 1445                              | Agrolandia         | 260              | 20          | 0.8                        | 416,000                   | 7,192,000  | 5.0                |
|                 |                |                                   | Pouso Redondo      | 2,045            | 20          | 0.8                        | 3,272,000                 |            |                    |
|                 |                |                                   | Agronomica         | 360              | 20          | 0.8                        | 576,000                   |            |                    |
|                 |                |                                   | Rio do Oeste       | 1,600            | 20          | 0.8                        | 2,560,000                 |            |                    |
|                 |                |                                   | Trombuto Central   | 80               | 20          | 0.8                        | 128,000                   |            |                    |
| Rio Do Sul      |                |                                   | 150                | 20               | 0.8         | 240,000                    |                           |            |                    |
| Itajai do Sul   | Barragem Sul   | 1273                              | Alfredo Wagner     | 155              | 20          | 0.8                        | 248,000                   | 248,000    | 0.2                |
|                 | Basin 4        | 381                               | Rio do Sul         | 150              | 20          | 0.8                        | 240,000                   | 240,000    | 0.6                |
| Itajai do Norte | Basin 5        | 1023                              | Ibirama            | 70               | 20          | 0.8                        | 112,000                   | 280,000    | 0.3                |
|                 |                |                                   | Presidente Getuilo | 65               | 20          | 0.8                        | 104,000                   |            |                    |
|                 |                |                                   | Vitor Meirelis     | 40               | 20          | 0.8                        | 64,000                    |            |                    |
| Benedito        | Rio Benedito   | 831                               | Benedito Novo      | 300              | 20          | 0.8                        | 480,000                   | 2,332,800  | 2.8                |
|                 |                |                                   | Dr. Pedrinho       | 808              | 20          | 0.8                        | 1,292,800                 |            |                    |
|                 |                |                                   | Timbo              | 350              | 20          | 0.8                        | 560,000                   |            |                    |
|                 | Rio dos Cedros | 600                               | Rio dos Cedros     | 1,100            | 20          | 0.8                        | 1,760,000                 | 2,320,000  | 3.9                |
|                 |                |                                   | Timbo              | 350              | 20          | 0.8                        | 560,000                   |            |                    |
| Itajai Acu      | Basin 6        | 906                               | Lontras            | 100              | 20          | 0.8                        | 160,000                   | 160,000    | 0.2                |
|                 | Basin 7        | 556                               | Ascurra            | 567              | 20          | 0.8                        | 907,200                   | 2,294,400  | 4.1                |
|                 |                |                                   | Indaial            | 250              | 20          | 0.8                        | 400,000                   |            |                    |
|                 |                |                                   | Rodeio             | 617              | 20          | 0.8                        | 987,200                   |            |                    |
|                 | Basin 9        | 349                               | Gaspar             | 3,400            | 20          | 0.8                        | 5,440,000                 | 5,440,000  | 15.6               |
|                 | Basin 10       | 227                               | Ilhota             | 3,000            | 20          | 0.8                        | 4,800,000                 | 4,800,000  | 21.1               |
|                 | Luiz Alves     | 580                               | Luis Alves         | 558              | 20          | 0.8                        | 892,800                   | 892,800    | 1.5                |
|                 | Basin12        | 176                               | Navegantes         | 1,200            | 20          | 0.8                        | 1,920,000                 | 3,840,000  | 21.8               |
|                 |                |                                   | Itajai             | 1,200            | 20          | 0.8                        | 1,920,000                 |            |                    |
| Itajai Mirim    | Basin11        | 472                               | Itajai             | 1,200            | 20          | 0.8                        | 1,920,000                 | 1,920,000  | 4.1                |
|                 | Itajai Mirim   | 1207                              | Brusque            | 170              | 0           | 0.8                        | 0                         | 0          | 0.0                |
| Sum             |                | 11596                             |                    | 26,295           |             |                            |                           | 41,800,000 |                    |

Source: JICA Survey Team

b. Change of the current dam operation method at the Sul dam

As the Sul dam has a larger storage capacity against the flood inflow compared to the Oeste dam, the inflow of 5-year flood is able to be fully stored in the reservoir with closure of all the gates. As shown in Figure 5.4.2, the stored volume is  $58.4 \times 10^6 \text{ m}^3$ , which is within its storage capacity  $93.5 \times 10^6 \text{ m}^3$ . According to DEINFRA, almost no overflowing from the spillway has occurred to date except the large floods in 1983 and 1984. As for the 5-year flood, the current dam operation method shall be changed to close all of the gates. In this connection, however, heightening of dam would be required because this change of dam operation method might increase the risk of overflowing through the spillway in times of large floods.



Remark: After the peak of flood, gates are opened in order when the flood inflow decreases to be  $180 \text{ m}^3/\text{s}$ , which is sufficiently smaller than the current flow capacity in the downstream river channel.

Source: JICA Survey Team

**Figure 5.4.2 Proposed Dam Operation against 5-year Flood at Sul Dam**

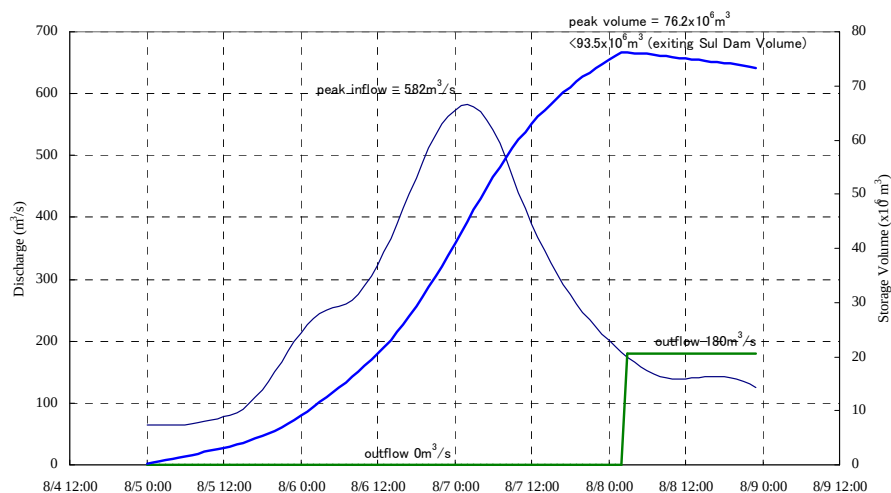
## (2) Basic Approach for Planning

Same as the flood control planning for the 5-year flood, priority is given to the alternative measures in river basin such as water storage in paddy fields and basin storage (small dam), followed by effective utilization of the existing flood control dams. As for the Itajai Mirim River and water storage in paddy field, the same measures for the 5-year flood are applied to the 10-year flood.

## (3) Flood Control Planning

### a. Change of the current dam operation method at the Sul dam

The inflow of 10-year flood is able to be fully stored in the reservoir with closure of all the gates at the Sul dam. As shown in Figure 5.4.8, the stored volume is  $76.2 \times 10^6 \text{ m}^3$ , which is within its storage capacity  $93.5 \times 10^6 \text{ m}^3$ . As mentioned in preceding sub-section, heightening of dam would be also required because this change of dam operation method might increase the risk of overflowing through the spillway in times of large floods.



Remark: After the peak of flood, gates are opened in order when the flood inflow decreases to be  $180 \text{ m}^3/\text{s}$ , which is sufficiently smaller than the current flow capacity in the downstream river channel.

Source: JICA Survey Team

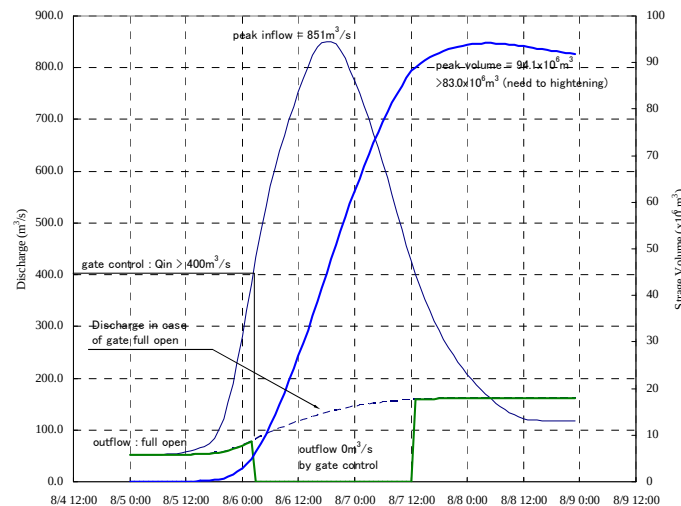
**Figure 5.4.8 Proposed Dam Operation against 10-year Flood at Sul Dam**

### b. Change of the current dam operation method at the Oeste dam

Overflowing from the spillway has occurred in several times to date at the Oeste dam, as the Oeste dam has a smaller storage capacity compared to the Sul dam. However, the 10-year flood is able to be controlled with full opening of gates when the flood inflow exceeds  $400 \text{ m}^3/\text{sec}$  to avoid inundation in Taio city located immediately downstream of the dam. As shown in Figure 5.4.9, the stored volume is  $72.2 \times 10^6 \text{ m}^3$ , which is within its storage capacity  $83.0 \times 10^6 \text{ m}^3$ . Same as the Sul dam, heightening of dam would be also required to reduce the increased risk of overflowing due to the change of dam operation method.

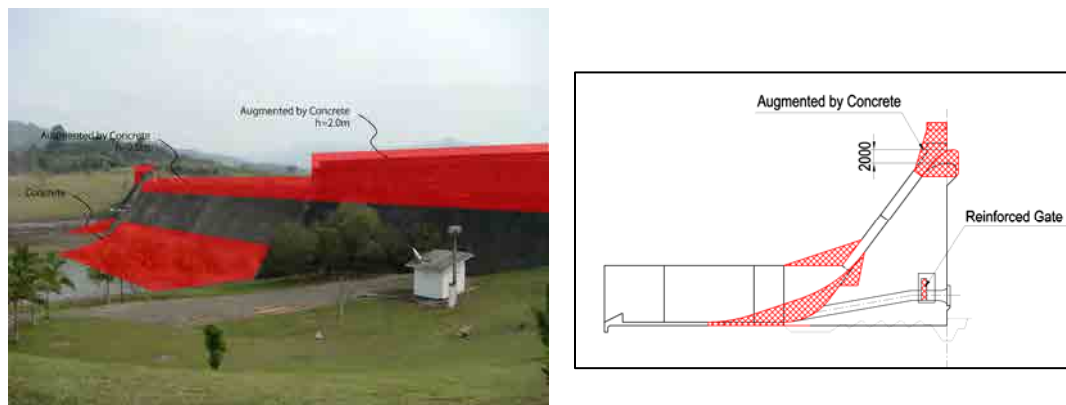
b. Heightening and change of the current dam operation method at the Oeste dam

The Oeste dam is proposed to be heightened by 2 m to increase flood control capacity. The capacity will be increased by approximately 16,200,000m<sup>3</sup>. Figure 5.4.21 shows the dam operation against the 25-year flood. When the flood inflow exceeds 400 m<sup>3</sup>/sec, all the gates shall be closed in order.



Source: JICA Survey Team

**Figure 5.4.21 Proposed Dam Operation against the 25-year Flood at Oeste Dam**



Source: JICA Survey Team

**Figure 5.4.22 Diagram of Heightening Oeste Dam**

c. River improvement in Taió and Timbó cities

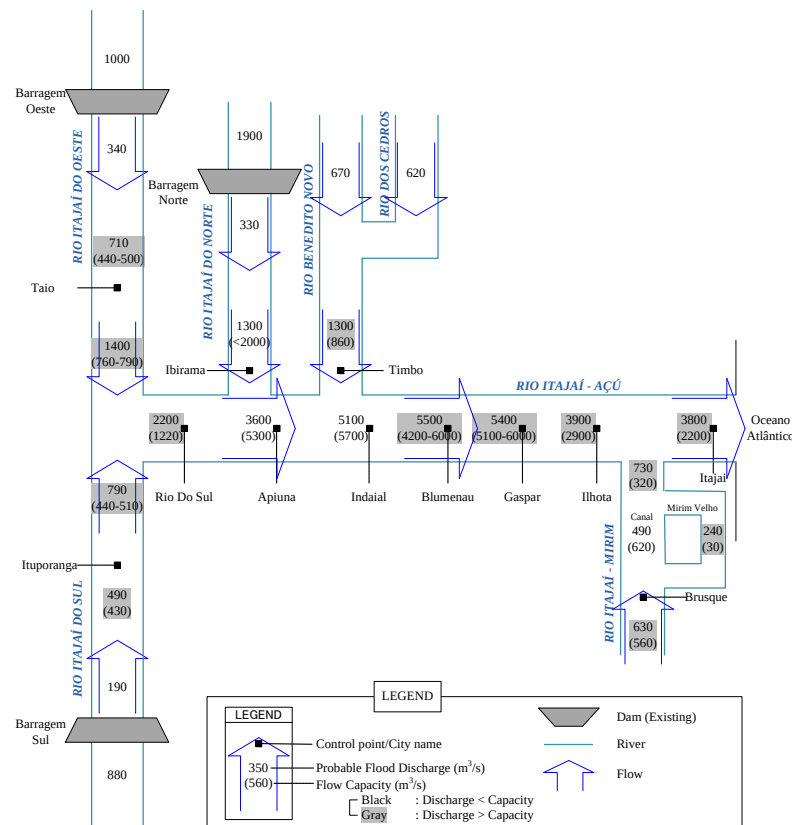
For both cities, river improvement is proposed considering that they are relatively small cities and river improvement might be possible without any relocation of residents. As for Taió city, river widening was proposed because of gentle riverbed slope. The current flow capacity would be increased 490 m<sup>3</sup>/s from 440 m<sup>3</sup>/s.

The Rio dos Cedros River joins the Benedit River in Timbó city. The current flow capacity downstream of the confluence is 860m<sup>3</sup>/s, which is smaller than the 25-year flood discharge of 1200 m<sup>3</sup>/s. Besides, the flow capacity of the Rio dos Cedros River in Timbó city is around 450 m<sup>3</sup>/s less than the 25-year flood discharge of 590m<sup>3</sup>/s. However, since only partial portions of river stretches have insufficient flow capacity, where the river bank elevation is relatively low, partial river improvement by embankment was planned as illustrated below.

cities along the Itajaí River having are smaller than the 50-year flood discharge.

## (2) Basic Approach for Planning

Even though river basin measures such as water storage in paddy fields, basin storage (small dam), effective utilization of the existing flood control dams are provided, it is difficult in Rio do Sul, Taió, Timbó Blumenau, and Itajaí cities to control the 50-year flood discharges smaller than the current flow capacity. Therefore, river improvement such as river widening, dyke and floodway are added to the alternatives for comparative study. In addition, a new flood control dam is considered to decrease the 50-year flood discharge in Brusque city.



Source: JICA Survey Team

**Figure 5.4.30 Distribution of 50-year Flood Discharge and Flow Capacity by City under Present Condition**

## (3) Flood Control Planning

Same as the proposed plan for the 25-year flood, water storage in paddy field, pre-releasing of two hydropower dams and 7 small dams are incorporated into the plan for the 50-year flood. Other flood control measures are outlined below.

### a. Heightening of the Sul dam

The crest of existing spillway is proposed to be heightened by 2 m. Due to the heightening, the storage capacity of the Sul dam would be increased to be 110 million m<sup>3</sup>. As shown in Figure 5.4.31, the 50-year flood is able to be controlled with full opening of gates when the flood inflow exceeds 300 m<sup>3</sup>/sec. The stored water volume will be 107.9 million m<sup>3</sup>.

## CHAPTER 7 FEASIBILITY STUDY FOR HEIGHTENING OF EXISTING FLOOD CONTROL DAMS AND MODIFICATION OF DAM OPERATION

### 7.1 Current Condition and Basic Policy for Heightening of Dams

#### 7.1.1 Necessity of Heightening Dams

There are three existing dams for flood control in the Itajai River basin, named the Norte Dam, the Oeste Dam, Sul Dam. Norte Dam has enough volume for flood control. On the other hand, as the reservoir volume of the Oeste Dam and Sul Dam is relatively small, floods have often overflowed from the spillway and brought some damage to the downstream area.

The M/P proposed the heightening of the Oeste Dam and Sul Dam and modification of their operations. The Oeste Dam and the spillway of Sul Dam are proposed to be heightened by 2 m. As the Sul Dam is of rockfill type, the dam body is difficult to be heightened.

#### 7.1.2 Existing Flood Control Dams

##### (1) General Features

Both of the Oeste and Sul Dams were constructed for the flood control purpose. These dams should be operated in empty condition in reservoir.

Figure 7.1.1 shows the location of dams. The general features of those dams are shown in Table 7.1.1. The Oeste Dam is of concrete gravity type. The Sul Dam is of Rockfill type, so the spillway is independent of the dam body.

**Table 7.1.1 General Features of the Oeste and Sul Dams**

|           |                                                     | <b>Oeste Dam</b>                       | <b>Sul Dam</b>                         |
|-----------|-----------------------------------------------------|----------------------------------------|----------------------------------------|
| Reservoir | Catchment area                                      | 1,042 km <sup>2</sup>                  | 1,273 km <sup>2</sup>                  |
|           | Reservoir area                                      | 950 ha                                 | 840 ha                                 |
|           | Total reservoir capacity                            | 83,000,000 m <sup>3</sup>              | 93,500,000 m <sup>3</sup>              |
|           | Minimum water level                                 | EL.340.0 m                             | EL.372.9 m                             |
|           | Maximum water level                                 | EL.362.5 m                             | EL.408.0 m                             |
| Dam body  | Year of completion                                  | 1973                                   | 1976                                   |
|           | Type of dam                                         | Concrete gravity                       | Rock-fill                              |
|           | Height of dam                                       | 25.0 m                                 | 43.5 m                                 |
|           | Crest length of dam                                 | 422 m                                  | 390 m                                  |
|           | Crest elevation of dam                              | 363.0 m                                | 410.0 m                                |
| Conduit   | Number of conduits                                  | 7                                      | 5                                      |
|           | Type of flood control                               | Gate control                           | Gate control                           |
|           | Diameter at gate                                    | 1500 mm                                | 1500 mm                                |
|           | Elevation of pipe center                            | 340.05 m                               | Around 368 m                           |
|           | Discharge capacity<br>(crest elevation of spillway) | 163 m <sup>3</sup> /s<br>(at WL.360 m) | 194 m <sup>3</sup> /s<br>(at WL.399 m) |
| Spillway  | Width                                               | Around 100m                            | 65m                                    |
|           | Crest elevation of spillway                         | 360.0m                                 | 399.0m                                 |
|           | Discharge capacity<br>(at maximum water level)      | Conduit 175 m <sup>3</sup> /s          | Conduit 217 m <sup>3</sup> /s          |
|           |                                                     | Spillway 1,125 m <sup>3</sup> /s       | Spillway 3,053 m <sup>3</sup> /s       |
|           |                                                     | Total 1,300 m <sup>3</sup> /s          | Total 3,270 m <sup>3</sup> /s          |

Source: Hatching: web site of DEINFR <http://www.deinfra.sc.gov.br/barragens/sobre-as-barragens/>

Discharge capacity : see Supporting Report A, Table 7.4.6

Others: Drawing not authorized, or hearing from the department of the SC State